Problem 9.11. Consider a particle of charge q and mass m, free to move in the xy plane in response to an electromagnetic wave propagating in the z direction (Eq. 9.49 – might as well set $\delta = 0$).

(a) Ignoring the magnetic force, find the velocity of the particle, as a function of time. (Assume the average velocity is zero.)
(b) Now calculate the resulting magnetic force on the particle.
(c) Show that the (time) average magnetic force is *zero*.

The problem with this naïve model for the pressure of light is that the velocity is $90°$ out of phase with the fields. For energy to be absorbed, there's got to be some *resistance* to the motion of the charges. Suppose we include a force of the form $-\gamma m\mathbf{v}$, for some damping constant γ.

(d) Repeat part (a) (ignore the exponentially damped transient). Repeat part (b), and find the average magnetic force on the particle.[12]

Problem 9.12. In the complex notation there is a clever device for finding the time average of a product. Suppose $f(\mathbf{r}, t) = A\cos(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta_a)$ and $g(\mathbf{r}, t) = B\cos(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta_b)$. Show that $\langle fg\rangle = (1/2)\,\mathrm{Re}(\tilde{f}\tilde{g}^*)$, where the star denotes complex conjugation. [Note that this only works if the two waves have the same $\mathbf{k}$ and ω, but they need not have the same amplitude or phase.] For example,

$$\langle u\rangle = \frac{1}{4}\,\mathrm{Re}\left(\epsilon_0\tilde{\mathbf{E}}\cdot\tilde{\mathbf{E}}^* + \frac{1}{\mu_0}\tilde{\mathbf{B}}\cdot\tilde{\mathbf{B}}^*\right) \quad \text{and} \quad \langle\mathbf{S}\rangle = \frac{1}{2\mu_0}\,\mathrm{Re}(\tilde{\mathbf{E}}\times\tilde{\mathbf{B}}^*).$$

Problem 9.13. Find all elements of the Maxwell stress tensor for a monochromatic plane wave traveling in the z direction and linearly polarized in the x direction (Eq. 9.49). Does your answer make sense? (Remember that $-\overleftrightarrow{\mathbf{T}}$ represents the momentum flux density.) How is the momentum flux density related to the energy density, in this case?

9.3 Electromagnetic Waves in Matter

9.3.1 Propagation in Linear Media

Inside matter, but in regions where there is no *free* charge or *free* current, Maxwell's equations become

$$\left.\begin{aligned} &\text{(i)}\ \ \boldsymbol{\nabla}\cdot\mathbf{D} = 0, \quad \text{(iii)}\ \ \boldsymbol{\nabla}\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}, \\ &\text{(ii)}\ \ \boldsymbol{\nabla}\cdot\mathbf{B} = 0, \quad \text{(iv)}\ \ \boldsymbol{\nabla}\times\mathbf{H} = \frac{\partial\mathbf{D}}{\partial t}. \end{aligned}\right\} \tag{9.66}$$

If the medium is *linear*,

$$\mathbf{D} = \epsilon\mathbf{E}, \quad \mathbf{H} = \frac{1}{\mu}\mathbf{B}, \tag{9.67}$$

[12] C. E. Mungan, *Am. J. Phys.* **77**, 965 (2009). See also Prob. 9.36.

and *homogeneous* (so ϵ and μ do not vary from point to point), they reduce to

$$\left.\begin{aligned} &\text{(i)}\ \ \boldsymbol{\nabla}\cdot\mathbf{E}=0, \quad \text{(iii)}\ \ \boldsymbol{\nabla}\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial t},\\ &\text{(ii)}\ \ \boldsymbol{\nabla}\cdot\mathbf{B}=0, \quad \text{(iv)}\ \ \boldsymbol{\nabla}\times\mathbf{B}=\mu\epsilon\frac{\partial\mathbf{E}}{\partial t}, \end{aligned}\right\} \tag{9.68}$$

which differ from the vacuum analogs (Eqs. 9.40) only in the replacement of $\mu_0\epsilon_0$ by $\mu\epsilon$.[13] Evidently, electromagnetic waves propagate through a linear homogeneous medium at a speed

$$v=\frac{1}{\sqrt{\epsilon\mu}}=\frac{c}{n}, \tag{9.69}$$

where

$$n\equiv\sqrt{\frac{\epsilon\mu}{\epsilon_0\mu_0}} \tag{9.70}$$

is the **index of refraction** of the substance. For most materials, μ is very close to μ_0, so

$$n\approx\sqrt{\epsilon_r}, \tag{9.71}$$

where ϵ_r is the dielectric constant[14] (Eq. 4.34). Since ϵ_r is almost always greater than 1, light travels *more slowly* through matter – a fact that is well known from optics.

All of our previous results carry over, with the simple transcription $\epsilon_0\to\epsilon$, $\mu_0\to\mu$, and hence $c\to v$. The energy density is[15]

$$u=\frac{1}{2}\left(\epsilon E^2+\frac{1}{\mu}B^2\right), \tag{9.72}$$

and the Poynting vector is

$$\mathbf{S}=\frac{1}{\mu}(\mathbf{E}\times\mathbf{B}). \tag{9.73}$$

For monochromatic plane waves, the frequency and wave number are related by $\omega=kv$ (Eq. 9.11), the amplitude of $\mathbf{B}$ is $1/v$ times the amplitude of $\mathbf{E}$ (Eq. 9.48), and the intensity[16] is

[13] This observation is mathematically pretty trivial, but the physical implications are extraordinary: As the wave passes through, the fields busily polarize and magnetize all the molecules, and the resulting (oscillating) dipoles create their own electric and magnetic fields. These combine with the original fields in such a way as to create a *single* wave with the same frequency but a different speed. This remarkable conspiracy (known in optics as the **Ewald–Oseen extinction theorem**) is responsible for the phenomenon of **transparency**. It is a distinctly *non*trivial consequence of linearity. For further discussion see M. B. James and D. J. Griffiths, *Am. J. Phys.* **60**, 309 (1992); H. Fearn, D. F. V. James, and P. W. Milonni, *Am. J. Phys.* **64**, 986 (1996); M. Mansuripur, *Opt. Photonics News* **9**, 50 (1998).

[14] The dielectric constant is "constant" in the sense of being independent of the *amplitude* of $\mathbf{E}$, but it may well depend on the *frequency*, as we shall see. Thus, for example, if you quote the (static) dielectric constant for water, from Table 4.2, you will conclude that the index of refraction is 8.9, which is *wildly* off, for visible light ($n=1.33$).

[15] See Prob. 8.27; refer to Section 4.4.3 for the precise *meaning* of "energy density," in the context of linear media.

[16] The *momentum* carried by an electromagnetic wave in matter is controversial. See, for example, S. M. Barnett, *Phys. Rev. Lett.* **104**, 070401 (2010).